

AURA MARKETING AI OPERATING SYSTEM

Complete Codebase Architectural Review & Master Blueprint (Version 1.0)

Target Audience: Claude / Technical Reviewer

Tech Stack: Next.js 19, FastAPI, MongoDB, Razorpay

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1. Executive Summary & Core Vision

Aura Marketing is an **AI-powered Digital Marketing Operating System** engineered to resolve tool fragmentation. It unifies SEO research, website auditing, competitor intelligence, AI copywriting, video synthesis, multi-channel ad campaigns, and social scheduling into an outcome-driven loop:

DISCOVER (SEO) → ANALYZE (Doctor) → DIAGNOSE (Top 10) → RECOMMEND → CREATE (Copy/Video) → PUBLISH → MEASURE

Core Positioning: 'Your AI marketing team is working on your business.'

2. Complete Project File Inventory

```
AI agent/
├── backend/
│   ├── main.py           # Core FastAPI App (Auth, Wallet, Image Edit, Audit, Analytics)
│   ├── marketing_doctor.py # AI Marketing Doctor Diagnostic Brain & Live HTTP Inspection
│   ├── seo_engine.py      # Top 10 Opportunity Engine, 'Why Not Top 10?', SEO Fix Plans
│   ├── campaigns.py       # Multi-Channel Campaign Builder, Ad Creatives, Social Calendar
│   ├── generator.py        # AI Video Generative Engine & Keyart Synthesizer
│   ├── database.py         # MongoDB Driver & InMemoryDatabase Fallback
│   └── auth.py             # JWT Token Authentication & Password Hashing
└── src/app/
    ├── marketing/page.tsx # AI Marketing OS Main UI (Glassmorphism + Bento Grid)
    ├── dashboard/page.tsx # AI Video Studio Dashboard & Render Console
    └── wallet/page.tsx     # Token Store & Razorpay Payment Integration
```

3. Detailed Backend Architecture & Endpoints

3.1 Core Engine (backend/main.py)

Entry point for FastAPI, handling CORS, JWT auth, Razorpay wallet transactions, AI image editing, web auditing, and competitor scraping.

- **Auth & Wallet:** POST /api/auth/register, POST /api/wallet/verify-payment
- **AI Image Suite:** POST /api/image/ai/edit (Remove BG, Cleanup, Outpaint via Clipdrop/Stability AI)
- **Auditing:** POST /api/webaudit/lighthouse & POST /api/competitor/analyze

3.2 AI Marketing Doctor Brain (backend/marketing_doctor.py)

Runs live HTTP requests to target URLs measuring response latency (ms), SSL status, HTTP status codes, title tags, H1 headers, and meta descriptions. Computes 6-category scores (SEO, Content, Social, Ads, Conversions, AI Search).

3.3 Top 10 Opportunity Engine (backend/seo_engine.py)

Identifies keywords trapped in positions #11–20 (Page 2). Executes POST /api/seo/why-not-top10 comparing target URLs against Top 10 SERP competitors on word count, H2/H3 density, and internal links.

3.4 Multi-Channel Campaigns (backend/campaigns.py)

Orchestrates campaigns under unique CAMP_XXXX IDs. Generates Google/Meta/LinkedIn ads, 5-day social media calendars, and 3-step email sequences.

4. Frontend UI/UX Architecture (src/app/marketing/page.tsx)

Built with a **Glassmorphism + Bento Grid** design system:

- **Visual Theme:** #030712 deep space dark background, translucent backdrop-blur-xl glass cards.
- **Single Accent Color:** #10b981 (Emerald) reserved for primary CTAs and key KPI metrics.
- **■K Command Bar:** Global AI search modal supporting natural language queries.
- **Bento Grid Rows:** Doctor Hero Scan (span 8) + Health Gauge (span 4), 6 Compact Outcome KPIs, AI Video Preview (span 7) + AI Chat Assistant (span 5), Organic Traffic Trend Chart (span 8) + Priority AI Actions (span 4).

5. Digital Marketing Industry Negatives Solved

Industry Negative	Aura Marketing Solution
Rising CAC & Lower ROAS	Top 10 Opportunity Engine captures quick-win Page 2 organic traffic without ad spend.
Algorithm Volatility	Live HTTP inspection monitors server latency, indexing headers & Core Web Vitals continuously.
Attribution Blindness	Unified CAMP_XXXX tracking container combines first-party server telemetry and lead matching.
Page 2 Traps (#11–20)	AI SERP Diagnostic benchmarks page word count, headings & links against Top 10 competitors.
Tool Fragmentation	Unifies SEO, auditing, copywriting, video generation, social scheduling, and analytics into 1 platform.

6. Prompts for Technical Review by Claude

1. **Architecture & Modular Design:** Evaluate separation of concerns between `marketing_doctor.py`, `seo_engine.py`, `campaigns.py`, and `main.py`.
2. **Live HTTP Scan Reliability:** Is `httpx` with 6s timeout and fallback regex parsing optimal for real-time site scans?
3. **First-Party Attribution Model:** How can we expand the `CAMP_XXXX` model to support multi-touch attribution (First Touch, Last Touch, W-Shaped)?
4. **Frontend Performance:** What React 19 / Next.js optimizations would you recommend for state management in `marketing/page.tsx`?